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Semiconductor device and method of forming graphene-coated core embedded within TIM

Abstract

A semiconductor device has a substrate and electrical component disposed over the substrate. The electrical component can be a semiconductor die, semiconductor package, surface mount device, RF component, discrete electrical device, or IPD. A TIM is deposited over the electrical component. The TIM has a core, such as Cu, covered by graphene. A heat sink is disposed over the TIM, electrical component, and substrate. The TIM is printed on the electrical component. The graphene is interconnected within the TIM to form a thermal path from a first surface of the TIM to a second surface of the TIM opposite the first surface of the TIM. The TIM has thermoset material or soldering type matrix and the core covered by graphene is embedded within the thermoset material or soldering type matrix. A metal layer can be formed between the TIM and electrical component.

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Background/Summary

FIELD OF THE INVENTION

(1) The present invention relates in general to semiconductor devices and, more particularly, to a semiconductor device and method of heat dissipation using graphene-coated core embedded within thermal interface material (TIM).

BACKGROUND OF THE INVENTION

(2) Semiconductor devices are commonly found in modern electronic products. Semiconductor devices perform a wide range of functions, such as signal processing, high-speed calculations, transmitting and receiving electromagnetic signals, controlling electronic devices, photo-electric, and creating visual images for television displays. Semiconductor devices are found in the fields of communications, power conversion, networks, computers, entertainment, and consumer products. Semiconductor devices are also found in military applications, aviation, automotive, industrial controllers, and office equipment.

(3) The SIP module includes high speed digital and RF electrical components, highly integrated for small size and low height, and operating at high clock frequencies and high power rating. The electrical components are known to generate substantial heat, which must be properly dissipated. Copper is good material to solderability and wettability of solder paste. A need still exists to improve heat dissipation, particularly in applications involving high speed digital and RF electrical components.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIGS. 1a-1c illustrate a semiconductor wafer with a plurality of semiconductor die separated by a saw street;

(2) FIGS. 2a-2g illustrate a process of forming a graphene Cu core within TIM for a SiP;

(3) FIGS. 3a-3e illustrate a process of forming a graphene Cu core;

(4) FIGS. 4a-4b illustrate further detail of forming the graphene Cu core within TIM; and

(5) FIG. 5 illustrates a printed circuit board (PCB) with different types of packages disposed on a surface of the PCB.

DETAILED DESCRIPTION OF THE DRAWINGS

(6) The present invention is described in one or more embodiments in the following description with reference to the figures, in which like numerals represent the same or similar elements. While the invention is described in terms of the best mode for achieving the invention's objectives, it will be appreciated by those skilled in the art that it is intended to cover alternatives, modifications, and equivalents as may be included within the spirit and scope of the invention as defined by the appended claims and their equivalents as supported by the following disclosure and drawings. The features shown in the figures are not necessarily drawn to scale. Elements having a similar function are assigned the same reference number in the figures. The term "semiconductor die" as used herein refers to both the singular and plural form of the words, and accordingly, can refer to both a single semiconductor device and multiple semiconductor devices.

(7) Semiconductor devices are generally manufactured using two complex manufacturing processes: front-end manufacturing and back-end manufacturing. Front-end manufacturing involves the formation of a plurality of die on the surface of a semiconductor wafer. Each die on the wafer contains active and passive electrical components, which are electrically connected to form functional electrical circuits. Active electrical components, such as transistors and diodes, have the ability to control the flow of electrical current. Passive electrical components, such as capacitors, inductors, and resistors, create a relationship between voltage and current necessary to perform electrical circuit functions.

(8) Back-end manufacturing refers to cutting or singulating the finished wafer into the individual semiconductor die and packaging the semiconductor die for structural support, electrical interconnect, and environmental isolation. To singulate the semiconductor die, the wafer is scored and broken along non-functional regions of the wafer called saw streets or scribes. The wafer is singulated using a laser cutting tool or saw blade. After singulation, the individual semiconductor die are disposed on a package substrate that includes pins or contact pads for interconnection with other system components. Contact pads formed over the semiconductor die are then connected to contact pads within the package. The electrical connections can be made with conductive layers, bumps, stud bumps, conductive paste, or wirebonds. An encapsulant or other molding material is deposited over the package to provide physical support and electrical isolation. The finished package is then inserted into an electrical system and the functionality of the semiconductor device is made available to the other system components.

(9) FIG. 1a shows a semiconductor wafer **100** with a base substrate material **102**, such as silicon, germanium, aluminum phosphide, aluminum arsenide, gallium arsenide, gallium nitride, indium phosphide, silicon carbide, or other bulk material for structural support. A plurality of semiconductor die or components **104** is formed on wafer **100** separated by a non-active, inter-die wafer area or saw street **106**. Saw street **106** provides cutting areas to singulate semiconductor wafer **100** into individual semiconductor die **104**. In one embodiment, semiconductor wafer **100** has a width or diameter of 100-450 millimeters (mm). Alternatively, wafer **100** can be a mold surface, organic or inorganic substrate, or target substrate suitable for graphene transfer.

(10) FIG. 1B shows a cross-sectional view of a portion of semiconductor wafer **100**. Each semiconductor die **104** has a back or non-active surface **108** and an active surface **110** containing analog or digital circuits implemented as active devices, passive devices, conductive layers, and dielectric layers formed within the die and electrically interconnected according to the electrical design and function of the die. For example, the circuit may include one or more transistors, diodes, and other circuit elements formed within active surface **110** to implement analog circuits or digital circuits, such as digital signal processor (DSP), application specific integrated circuits (ASIC), memory, or other signal processing circuit. Semiconductor die **104** may also contain IPDs, such as inductors, capacitors, and resistors, for RF signal processing.

(11) An electrically conductive layer **112** is formed over active surface **110** using physical vapor deposition (PVD), chemical vapor deposition (CVD), electrolytic plating, electroless plating process, or other suitable metal deposition process. Conductive layer **112** can be one or more layers of aluminum (Al), copper (Cu), tin (Sn), nickel (Ni), gold (Au), silver (Ag), or other suitable electrically conductive material. Conductive layer **112** operates as contact pads electrically connected to the circuits on active surface **110**.

(12) An electrically conductive bump material is deposited over conductive layer **112** using an evaporation, electrolytic plating, electroless plating, ball drop, or screen printing process. The bump material can be Al, Sn, Ni, Au, Ag, Pb, Bi, Cu, solder, and combinations thereof, with an optional flux solution. For example, the bump material can be eutectic Sn/Pb, high-lead solder, or lead-free solder. The bump material is bonded to conductive layer **112** using a suitable attachment or bonding process. In one embodiment, the bump material is reflowed by heating the material above its melting point to form balls or bumps **114**. In one embodiment, bump **114** is formed over an under bump metallization (UBM) having a wetting layer, barrier layer, and adhesive layer. Bump **114** can also be compression bonded or thermocompression bonded to conductive layer **112**. Bump **114** represents one type of interconnect structure that can be formed over conductive layer **112**. The interconnect structure can also use bond wires, conductive paste, stud bump, micro bump, or other electrical interconnect.

(13) FIGS. 2a-2g illustrate a process of forming an SiP module with graphene with the TIM over electrical component for thermal dissipation. FIG. 2a shows a cross-sectional view of multi-layered interconnect substrate **120** including conductive layers **122** and insulating layer **124**. Conductive

layer **122** can be one or more layers of Al, Cu, Sn, Ni, Au, Ag, or other suitable electrically conductive material. Conductive layers can be formed using PVD, CVD, electrolytic plating, electroless plating process, or other suitable metal deposition process. Conductive layer **122** provides horizontal electrical interconnect across substrate **120** and vertical electrical interconnect between top surface **126** and bottom surface **128** of substrate **120**. Portions of conductive layer **122** can be electrically common or electrically isolated depending on the design and function of semiconductor die **104** and other electrical components. Insulating layer **124** contains one or more layers of silicon dioxide (SiO₂), silicon nitride (Si₃N₄), silicon oxynitride (SiON), tantalum pentoxide (Ta₂O₅), aluminum oxide (Al₂O₃), solder resist, polyimide, benzocyclobutene (BCB), polybenzoxazoles (PBO), and other material having similar insulating and structural properties. Insulating layers can be formed using PVD, CVD, printing, lamination, spin coating, spray coating, sintering or thermal oxidation. Insulating layer **124** provides isolation between conductive layers **122**.

(14) In FIG. **2b**, electrical components **130a-130c** are disposed on surface **126** of interconnect substrate **120** and electrically and mechanically connected to conductive layers **122**. Electrical components **130a-130c** are positioned over substrate **120** using a pick and place operation. For example, electrical components **130a** and **130b** can be discrete electrical devices, or IPDs, such as a diode, transistor, resistor, capacitor, and inductor. Electrical component **130c** can be semiconductor die **104** from FIG. **1c** with bumps **114** oriented toward surface **126** of substrate **120**. Alternatively, electrical components **130a-130c** can include other semiconductor die, semiconductor packages, surface mount devices, RF components, discrete electrical devices, or integrated passive devices (IPD). FIG. **2c** illustrates electrical components **130a-130c** electrically and mechanically connected to conductive layers **122** of substrate **120**. Conductive paste or solder **136** provides electrical and mechanical connection to terminals **132** and **134** of electrical components **130a** and **130b**, respectively. Bumps **114** provide electrical and mechanical connection for electrical component **130c**.

(15) In FIG. **2d**, TIM **140** is deposited over surface **108** of electrical component **130c**. Alternatively, metal coating **141** is first deposited over surface **108** of electrical component **130c**, depending on the type of TIM. Metal coating **141** can be Ti, Ag, or SUS/Cu. TIM **140** is deposited over metal coating **141**. In one embodiment, TIM **140** is printed on surface **108** of electrical component **130c** using a 3D printer. FIG. **3a** illustrates electrical component **130c** disposed on printer bed **142**. Printer bed **142** is heated to 80-100° C. TIM **140** is dispensed from printer nozzle **144** onto surface **108** of electrical component **130c**. Printer bed **142** moves three dimensionally (x, y, z directions) to control distribution of TIM **140** on surface **108**.

(16) FIG. **3b** illustrates further detail of TIM **140** including a plurality of cores **150** surrounded or covered by graphene **152** embedded in matrix **154**. In one embodiment, matrix **154** is a thermoset material, such epoxy resin or adhesive with filler containing alumina, Al, aluminum zinc oxide, or other material having good heat transfer properties. Matrix **154** can be thermal grease such as silicon or polymer type such as polymethyl methacrylate (PMMA) or polyethylene terephthalate (PET). FIG. **3c** illustrates core **150**. In one embodiment, core **150** is Cu, Ni, phase change material (PCM), or other suitable metal or similar material.

(17) FIG. **3d** illustrates graphene coating **152** formed around surface **151** of metal core **150**. FIG. **3e** illustrates further detail of graphene coating **152** formed as a mesh network around surface **151** of metal core **150**, collectively graphene Cu core **156**. Graphene **152** is an allotrope of carbon with one or more layers of carbon atoms each arranged in a two-dimensional (2D) honeycomb lattice. Graphene **152** can be formed by CVD. Metal core **150** is placed in a chamber heated to 900-1080° C. A gas mixture of CH₄/H₂/Ar is introduced into the chamber to initiate a CVD reaction. The carbon source decomposes in the high-temperature reaction chamber as the CVD reaction separates the carbon atoms from the hydrogen atoms, leaving graphene **152** on surface **151** of metal core **150**. The release of carbon atoms over metal core **150** forms continuous sheet of graphene

152. Additional information related to forming graphene by CVD is disclosed in U.S. Pat. No. 8,535,553, and hereby incorporated by reference.

(18) In another embodiment, matrix **154** is a polymer with dispersed graphene, carbon nanotubes, conductive polymers, and the like. Core **150** is PCM capable of phase change from solid to liquid phase or from liquid phase to solid phase within the operating temperature range of the semiconductor chip, e.g., 20-200° C. A first coating **152** is formed around PCM core **150**, and a second coating **153** is formed between the first coating **152** and PCM core **150**, as shown in FIG. **3d** and discussed in published Korean application KR101465616B1. Second coating **153** is a polymer intermediate layer. Matrix **154** with graphene covered core is further disclosed in U.S. patent Ser. No. 10/421,123, and hereby incorporated by reference. Matrix **154** with graphene covered core offers high thermal transfer.

(19) The properties of graphene are summarized in Table 1, as follows:

(20) TABLE-US-00001 TABLE 1 Properties of graphene
Parameter Electronic mobility $2 \times 10^{sup.5}$ cm.^{sup.2} V.^{sup.-1} s.^{sup.-1} Current density $10^{sup.9}$ A cm.^{sup.-1} Velocity of fermion (electron) $10^{sup.6}$ m s.^{sup.-1} Thermal conductivity 4000-5000 W m.^{sup.-1} K.^{sup.-1} Tensile strength 1.5 Tpa Breaking strength 42N m.^{sup.-1} Transparency 97.7% Elastic limit 20% Surface area 2360 m.^{sup.2} g.^{sup.-1}

(21) Graphene **152** exhibits high thermal conductivity. A plurality of graphene Cu cores **156** physically interconnects within thermoset material **154**, as shown in FIG. **3b**, to create a thermal path **158** between surface **160** and surface **162** of TIM **140**. Heat from electrical component **130c** is dissipated from surface **108** and surface **160** through thermal path **158** by way of connecting graphene Cu cores **156** to surface **162**.

(22) In FIG. **2e**, heat sink or heat spreader **170** is disposed over electrical components **130a-130c**, including TIM **140** deposited over electrical component **130c**. Heat sink **170** can be one or more layers of Al, Cu, Sn, Ni, Au, Ag, or other suitable thermally conductive material. FIG. **2f** illustrates heat sink **170** mounted to substrate **120** with paste **172** and cure. Heat sink **170** dissipates heat generated by electrical components **130a-130c**, as transferred through graphene Cu core **156** and TIM **140** to the heat sink. Heat sink **170** may include extensions or tabs **174** extending vertical or perpendicular with respect to surface **176** of the heat sink, as in FIG. **2g**. Extensions **174** provide additional surface area for heat dissipation.

(23) FIG. **4a** illustrates further detail within region **180** from FIG. **2f** with TIM **140** deposited over electrical component **130c** and heat sink **170** disposed over the TIM. TIM **140** including a plurality of metal cores **150** surrounded by graphene **152** embedded in thermoset material **154**, see FIGS. **3a-3d**. A plurality of graphene Cu cores **156** physically connects within thermoset material **154** to create a thermal path **158** including portions of interconnected graphene **152** on adjacent metal cores **150** disposed and extending between surface **160** and surface **162** of TIM **140**. Heat from electrical component **130c** is dissipated from surface **108** and surface **160** through thermal path **158** by way of connecting graphene Cu cores **156** to surface **162**.

(24) FIG. **4b** illustrates another embodiment within region **180** with TIM **140** deposited over electrical component **130c** and heat sink **170** disposed over the TIM. TIM **140** including a plurality of metal cores **150** surrounded by graphene **152** embedded in soldering type matrix **182**. Soldering type matrix **182** can be indium (In) or InAg with a high thermal conductivity. In this case, metal coating **141** is formed over surface **108** of electrical component **130c**, and soldering type matrix **182** is deposited over metal coating **141**. A plurality of graphene Cu cores **156** physically connects within soldering type matrix **182** to create a thermal path **184** including portions of interconnected graphene **152** on adjacent metal cores **150** disposed and extending between surface **186** and surface **188** of TIM **140**. Heat from electrical component **130c** is dissipated from surface **108** and surface **186** through thermal path **184** by way of connecting graphene Cu cores **156** to surface **188**.

(25) The combination of interconnect substrate **120**, electrical components **130a-130c**, TIM **144** with graphene Cu core **156**, and heat sink **170** constitute SiP **200**. Graphene Cu core **156** aids with

the heat transfer capability of SiP **200**, particularly between electrical components **130a-130c**, known to generate heat, and heat sink **150**, useful to dissipate heat. Graphene **152** has a low moisture permeability and a high thermal conductivity of 4000-5000 W m.sup.-1 K.sup.-1, 10 times higher than Cu at room temperature. Since carbon also has a good solderability and wettability of solder paste, TIM **140** and heat sink **170** can be readily attached. Graphene **152** exhibits a high degree of flexibility and remains stable against warpage. Graphene **152** reduces or prevents oxidation. TIM **140** with graphene Cu core **156** improves thermal conductivity, while lowering manufacturing cost.

(26) FIG. 5 illustrates electrical device **400** having a chip carrier substrate or PCB **402** with a plurality of semiconductor packages disposed on a surface of PCB **402**, including SiP **200**. Electrical device **400** can have one type of semiconductor package, or multiple types of semiconductor packages, depending on the application.

(27) Electrical device **400** can be a stand-alone system that uses the semiconductor packages to perform one or more electrical functions. Alternatively, electrical device **400** can be a subcomponent of a larger system. For example, electrical device **400** can be part of a tablet, cellular phone, digital camera, communication system, or other electrical device. Alternatively, electrical device **400** can be a graphics card, network interface card, or other signal processing card that can be inserted into a computer. The semiconductor package can include microprocessors, memories, ASIC, logic circuits, analog circuits, RF circuits, discrete devices, or other semiconductor die or electrical components. Miniaturization and weight reduction are essential for the products to be accepted by the market. The distance between semiconductor devices may be decreased to achieve higher density.

(28) In FIG. 5, PCB **402** provides a general substrate for structural support and electrical interconnect of the semiconductor packages disposed on the PCB. Conductive signal traces **404** are formed over a surface or within layers of PCB **402** using evaporation, electrolytic plating, electroless plating, screen printing, or other suitable metal deposition process. Signal traces **404** provide for electrical communication between each of the semiconductor packages, mounted components, and other external system components. Traces **404** also provide power and ground connections to each of the semiconductor packages.

(29) In some embodiments, a semiconductor device has two packaging levels. First level packaging is a technique for mechanically and electrically attaching the semiconductor die to an intermediate substrate. Second level packaging involves mechanically and electrically attaching the intermediate substrate to the PCB. In other embodiments, a semiconductor device may only have the first level packaging where the die is mechanically and electrically disposed directly on the PCB. For the purpose of illustration, several types of first level packaging, including bond wire package **406** and flipchip **408**, are shown on PCB **402**. Additionally, several types of second level packaging, including ball grid array (BGA) **410**, bump chip carrier (BCC) **412**, land grid array (LGA) **416**, multi-chip module (MCM) or SIP module **418**, quad flat non-leaded package (QFN) **420**, quad flat package **422**, embedded wafer level ball grid array (eWLB) **424**, and wafer level chip scale package (WLCSP) **426** are shown disposed on PCB **402**. In one embodiment, eWLB **424** is a fan-out wafer level package (Fo-WLP) and WLCSP **426** is a fan-in wafer level package (Fi-WLP). Depending upon the system requirements, any combination of semiconductor packages, configured with any combination of first and second level packaging styles, as well as other electrical components, can be connected to PCB **402**. In some embodiments, electrical device **400** includes a single attached semiconductor package, while other embodiments call for multiple interconnected packages. By combining one or more semiconductor packages over a single substrate, manufacturers can incorporate pre-made components into electrical devices and systems. Because the semiconductor packages include sophisticated functionality, electrical devices can be manufactured using less expensive components and a streamlined manufacturing process. The resulting devices are less likely to fail and less expensive to manufacture resulting in a lower cost

for consumers.

(30) While one or more embodiments of the present invention have been illustrated in detail, the skilled artisan will appreciate that modifications and adaptations to those embodiments may be made without departing from the scope of the present invention as set forth in the following claims.

Claims

1. A semiconductor device, comprising: a substrate; an electrical component disposed over the substrate; a thermal interface material (TIM) deposited over the electrical component, wherein the TIM includes a plurality of cores covered by graphene and the graphene is formed as a mesh network in a two-dimensional honeycomb lattice over the cores and the graphene over the cores is in contact continuously between a first surface of the TIM to a second surface of the TIM opposite the first surface of the TIM to form a thermal path through the continuous contact of the graphene over the cores within the TIM between the first surface of the TIM and the second surface of the TIM; and a heat sink disposed over the TIM.
2. The semiconductor device of claim 1, wherein the plurality of cores includes copper.
3. The semiconductor device of claim 1, wherein the TIM is printed on the electrical component.
4. The semiconductor device of claim 1, wherein the TIM includes thermoset material or soldering type matrix and the plurality of cores covered by graphene is embedded within the thermoset material or soldering type matrix.
5. The semiconductor device of claim 1, further including a metal layer between the TIM and electrical component.
6. A semiconductor device, comprising: an electrical component; and a thermal interface material (TIM) deposited over the electrical component, wherein the TIM includes a plurality of cores covered by graphene and the graphene over the cores is in continuous contact between a first surface of the TIM to a second surface of the TIM opposite the first surface of the TIM to form a thermal path through the continuous contact of the graphene over the cores within the TIM.
7. The semiconductor device of claim 6, further including: a substrate, wherein the electrical component is disposed over the substrate; and a heat sink disposed over the TIM.
8. The semiconductor device of claim 6, wherein the plurality of cores includes copper.
9. The semiconductor device of claim 6, wherein the TIM is printed on the electrical component.
10. The semiconductor device of claim 6, wherein the TIM includes thermoset material or soldering type matrix and the plurality of cores covered by graphene is embedded within the thermoset material or soldering type matrix.
11. The semiconductor device of claim 6, further including a metal layer between the TIM and electrical component.
12. The device of claim 6, wherein the graphene is formed as a mesh network in a two-dimensional honeycomb lattice over the cores.
13. A method of making a semiconductor device, comprising: providing a substrate; disposing an electrical component over the substrate; depositing a thermal interface material (TIM) over the electrical component, wherein the TIM includes a plurality of cores covered by graphene and the graphene over the cores is in contact continuously between a first surface of the TIM to a second surface of the TIM opposite the first surface of the TIM to form a thermal path through the continuous contact of the graphene over the cores within the TIM; and disposing a heat sink over the TIM.
14. The method of claim 13, wherein the plurality of cores includes copper.
15. The method of claim 13, further including printing the TIM on the electrical component.
16. The method of claim 13, wherein the TIM includes thermoset material or soldering type matrix and the plurality of cores covered by graphene is embedded within the thermoset material or soldering type matrix.

17. The method of claim 13, further including forming a metal layer between the TIM and electrical component.

18. The method of claim 13, wherein the graphene is formed as a mesh network in a two-dimensional honeycomb lattice over the cores.

19. A method of making a semiconductor device, comprising: providing an electrical component; and depositing a thermal interface material (TIM) over the electrical component, wherein the TIM includes a plurality of cores covered by graphene and the graphene over the cores is in continuous contact between a first surface of the TIM to a second surface of the TIM opposite the first surface of the TIM to form a thermal path through the continuous contact of the graphene over the cores within the TIM to dissipate heat from the electrical component.

20. The method of claim 19, further including: providing a substrate, wherein the electrical component is disposed over the substrate; and disposing a heat sink over the TIM.

21. The method of claim 19, wherein the plurality of cores includes copper.

22. The method of claim 19, further including printing the TIM on the electrical component.

23. The method of claim 19, wherein the TIM includes thermoset material or soldering type matrix and the plurality of cores covered by graphene is embedded within the thermoset material or soldering type matrix.

24. The method of claim 19, wherein the graphene is formed as a mesh network in a two-dimensional honeycomb lattice over the cores.
